Phishing Email Detection Using Machine Learning

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Abstract*— Phishing email attacks remain one of the most widespread and damaging forms of cyberattacks, exploiting users to reveal sensitive information or interact with malicious content. As phishing techniques continue to evolve, traditional detection methods have become increasingly ineffective at identifying newer and more advanced threats. This project investigates the use of machine learning and deep learning models to automate and improve the detection of phishing emails. Using a dataset of over 29,000 labeled emails, classical machine learning models such as Logistic Regression, Support Vector Machines, Multinomial Naive Bayes, Random Forest, and K-Nearest Neighbors were evaluated based on their accuracy and evaluation time, along with deep learning models such as Long Short-Term Memory and Gated Recurrent Unit. This project compares each model's performance and highlights its strengths and limitations. The results showed that SVM had the highest accuracy of 99.11%, KNN had the lowest accuracy of 97.55%, and that classical machine learning models were able to achieve a similar result in a shorter amount of time compared to deep learning models. The findings of this project demonstrate the effectiveness of data-driven methods for phishing detection and provide insight into which models are best suited for real-world cybersecurity applications.

***Keywords*— phishing email detection, machine learning, deep learning, classification**

I. Introduction

Emails are one of the most widely used forms of communication in both personal and professional environments. Due to its accessibility, speed, and global reach, it has become a primary medium for information exchange. However, this widespread usage has also made emails one of the most common targets for cyberattacks. Among these attacks, phishing remains one of the most prevalent and dangerous threats. According to the Anti-Phishing Working Group's 2nd quarterly report for 2025, there have been 1,130,393 phishing attacks observed, with 18.2% of the attacks occurring in the Webmail sector [19].

Phishing emails are malicious messages designed to deceive users into revealing sensitive information such as passwords, financial information, or identification details. These attacks impersonate trusted organizations and individuals, making them difficult for users to detect. Phishing emails often attempt to distribute malware through harmful links or file attachments. As these techniques are becoming continuously refined, phishing emails are becoming more sophisticated and convincing. Traditional email filtering systems typically rely on rule-based approaches, predefined signatures, and blacklists, and while these methods are effective against known threats, they struggle to keep up with new and evolving phishing strategies.

To address these challenges, machine learning and deep learning techniques have emerged as powerful alternatives for phishing detection. These methods allow systems to learn directly from large datasets of labeled emails and automatically identify patterns associated with malicious behavior. By analyzing the content and structure of emails, machine learning and deep learning models are capable of providing faster, more adaptive, and more accurate detection compared to traditional methods. This project explores and evaluates multiple machine learning and deep learning models for phishing email classification, with the goal of identifying the model and technique best suited for this problem.

A. Motivation

Phishing attacks pose a serious threat to the safety of individuals, businesses, and organizations worldwide. Successful phishing attacks can lead to compromised accounts, stolen identities, and significant financial loss. As online services continue to expand, the potential impact of phishing attacks continues to grow.

Due to the widespread usage of emails across the globe, manual inspection and filtering systems are not viable and practical solutions for detecting phishing attempts. Automated detection systems that can operate on larger scales and respond to threats in real time must be implemented within online services. Machine learning and deep learning models provide the foundation for such automated systems due to their ability to distinguish between legitimate and malicious emails based on patterns in the data.

The major motivation of this project is to compare the effectiveness of different modeling approaches for phishing detection. Classical machine learning models, such as Logistic Regression, Support Vector Machines, and Naive Bayes, are known for their fast training times and strong performance on text classification tasks. On the other hand, deep learning models such as Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) are capable of capturing deeper linguistic and sequential patterns within email text. This allows them to identify complex phishing strategies that simpler models may fail to capture. By evaluating and comparing both machine learning and deep learning approaches, this project aims to contribute to the development of more effective cybersecurity solutions that can better protect users from modern phishing threats.

II. Literature Review

A. Classical Machine Learning

The decisions made in designing and developing this project were largely inspired by previous research and implementations of machine learning to the problem of phishing email detection. Al-Subaiey et al. [1] used SVM, Multinomial NB, and Random Forest as their machine learning models, and they compared the performances of the models when applying different NLP techniques: TF-IDF and Word2Vec. They found that SVM with TF-IDF had the highest accuracy of 99.4%.

Al Tawil et al. [5] used Logistic Regression, Random Forest, Decision Tree, and Multilayer Perceptron as their models, as well as the BERT LLM for comparison. They applied TF-IDE and Word2Vec to each machine learning model. MLP was found to have the highest accuracy for both NLP techniques, but BERT had the highest overall. Though BERT had the highest accuracy, it took significantly longer to train and evaluate compared to the machine learning models.

Fares et al. [7] used Random Forest, Decision Tree, and XGBoost, and applied both TF-IDF and Bag of Words to each model. They found that SVM had the highest accuracy of 97% while also having the longest training and prediction times.

Rawal et al. [8] extracted link, tag, and word-based features from a dataset of emails. They used SVM, Naive Bayes, Random Forest, Logistic Regression, and Voted Perceptron as their models, and found that SVM and Random Forest had the highest accuracies and performed equally well.

Harikrishnan et al. [11] used a variety of machine learning models and split the dataset into training, validation, and testing sets. TF-IDF was applied to all sets, while SVD and NMF, techniques for feature selection and dimensionality reduction, were applied to the validation and testing sets. They found that for the validation set, Random Forest, AdaBoost, and SVM had the highest accuracies.

In most of these studies, SVM and Random Forest had the best performance in terms of accuracy and evaluation speed, which were key factors in our decision to use them in our project. Though Logistic Regression and Naive Bayes also had good results, we decided to use Logistic Regression as a baseline for comparison and Naive Bayes as an alternative.

B. Deep Learning

Deep learning was applied in this project to compare the performance of its models to those of machine learning models. While machine learning models excel at fast and adaptive detection, the layered structure of deep learning models allows them to capture deeper linguistic patterns. In addition, deep learning models have the ability to interpret large and complex datasets without supplemental feature engineering. Using the findings of previous research, we selected LSTM and GRU as the deep learning models for our project.

Brindha et al. [12] developed a deep learning model that applied a Gated Recurrent Unit (GRU) classifier, which is a type of RNN that the authors consider to be a simplified model of LSTM. This is because GRU continuously reduces the number of gates in an LSTM model while ensuring that the long-term memories are taken into account [20]. GRU is an appropriate classifier for detection systems since it uses historical information to identify and analyze patterns, a key step in its classification process [20]. The model developed by [12] achieved an accuracy of 99.72%, which was the highest among other models such as LSTM and CNN. The GRU classifier's method of operating as well as its performance in previous studies were key factors in our decision to apply it in our project. Additionally, because it is closely related to LSTM, we decided to use LSTM as a baseline for comparison.

Other works experimented with several other deep learning models. Alhogail et al. [10] used a Graph Convolutional Network (GCN) classifier, which is a type of convolutional neural network that classifies nodes using semi-supervised learning methods. The model achieved an accuracy of 98.2%, which the authors attribute to their use of the email's body text as their only feature. Patra et al. [14] used BERT and Dense Passage Retrieval (DPR) as their deep learning models and achieved a 98.43% accuracy. Atawneh et al. [18] applied CNN, RNN, LSTM, and BERT as the deep learning models and found that BERT had the highest accuracy of 99.61%.

III. System Model and Background

The machine learning and deep learning models used the content of the emails as input, and the evaluation metrics for each model were considered as their outputs. The dataset used for this project was the Enron dataset, which was a part of the Phishing Email Dataset from Kaggle. Al-Subaiey et. al [1] created the Phishing Email Dataset by cleaning and combining six different email datasets, one of them being the Enron dataset. The steps taken in data cleaning were removal of punctuations, tokenization, removal of stopwords, and lemmatization. Before the data was processed into the models, it was converted to numerical matrices using TF-IDF.

For the machine learning models, the following algorithms were used for training and testing the data: Logistic Regression, Support Vector Machine, Multinomial Naive Bayes, Random Forest Classifier, and K-Nearest Neighbor. These models were selected based on previous research discussed in the literature review. The dataset was split into training and testing sets. After the sets were processed by each model, a table of evaluation metrics and a confusion matrix were produced. The evaluation metrics included accuracy, precision, F1-score, recall, and support. The models were evaluated based on the test sets.

For the deep learning models, Long Short-Term Memory and Gated Recurrent Unit were the algorithms used. Like the machine learning models, these models were selected based on results from previous research. Each model included an embedding layer followed by the deep learning algorithm, a dense layer with the ReLU activation function, dropout, and a dense layer with the sigmoid activation function. The models were compiled with the Adam optimizer and a binary cross-entropy loss function. The dataset was split into training, validation, and testing sets. The output of these models included evaluation metrics such as accuracy, precision, and recall. The models were evaluated based on the test sets.

The best machine learning and deep learning models were selected based on their overall performance, using accuracy as the primary metric and the other scores as supporting metrics. Determining whether machine learning or deep learning was a better fit for phishing detection was based on comparing these evaluation metrics, as well as comparing other factors such as speed and use cases.

This project was developed on Google Colab, using Pandas for storing and manipulating data and Scikit-learn for importing classical machine learning algorithms and evaluation metrics. TensorFlow and Keras were used for importing deep learning algorithms and developing the models. Matplotlib and Seaborn were used for visualizing the results as confusion matrices and WordClouds. RegEx and NLTK were additional libraries used for data preprocessing.

IV. Methodology

The Enron dataset from Kaggle was explored, and it was discovered to contain 29,797 emails, with 15,971 non-phishing (0) and the remaining 13,826 labeled as phishing (1). Since the dataset was divided into three columns, “subject”, “body”, and “label”, the “subject” and “body” columns were combined into one column for data processing. This column containing all the email content was then preprocessed by cleaning (or removing punctuation), tokenizing, removing stopwords, and lemmatizing the data. The data preprocessing was accomplished by utilizing the NLTK library. After cleaning the data, it was further explored by creating word clouds out of the email content under each label.

Then, the data was split into 20% for testing and 80% for training. The NLP (Natural Language Processing) technique TF-IDF (Term Frequency-Inverse Document Frequency) was used to create features for the classic machine learning models to train on. The classic machine learning classifiers used for this paper are Logistic Regression, SVM (Support Vector Machine), Multinomial NB (Naive Bayes), RFC (Random Forest Classifier), and KNN (K-Nearest Neighbors), made possible by importing the Sci-kit learn library.

Afterwards, two deep learning models, RNN (Recurrent Neural Networks), were trained using the TensorFlow library, which were LSTM (Long Short Term Memory) and GRU (Gated Recurrent Unit) classifiers. For these models, the data was split into 80% for training, 10% for validation, and the remaining 10% for testing. The data was then put into a text preprocessing layer from the Keras library before being trained in the deep learning models. When training the models, a batch size of 32 and a learning rate of 0.001 were used, and each model was trained for 5 epochs each for fair analysis.

V. Results

Before training the machine learning models, word clouds were made as visual representations of what the top 100 words of emails under each label (non-phishing and phishing) were. At first glance, the relevant words to the company Enron were common to the non-phishing cloud (Fig. 1), such as “enron” and “j kaminski.” In the phishing cloud (Fig. 2), less relevant words were used, and instead, more generic words were common, such as “email” and “united state.”

Fig. 1. Word Cloud of Non-Phishing Emails

Fig. 2. Word Cloud of Phishing Emails

After training the models, the accuracy, precision, recall, and F1-score of each were recorded for analysis. For the classic ML classifiers, the SVM (Support Vector Machine) model had the highest accuracy of 0.9911, and the KNN model had the lowest accuracy of 0.9755. After the SVM model, the classic models ranked in terms of accuracy in descending order would be RFC (Random Forest Classifier), NB (Naive Bayes), and LR (Logistic Regression).

As shown in Fig. 3, the results of the recurrent neural network (RNN) deep learning models, GRU (Gated Recurrent Unit) and LSTM (Long Short-Term Memory), are from evaluating the testing data. The GRU model has a higher accuracy than the LSTM model for the deep learning models. The GRU model has the second-highest accuracy compared to all of the models trained, as shown in Fig. 3.

The F1-score was manually calculated by hand as the harmonic mean of the precision and recall for the deep learning models. The GRU model also has a higher F1-score than the LSTM model, which suggests that the GRU model is a better model overall for detecting phishing.

In descending order of all of the models ranked in terms of accuracy, it would be SVM, GRU, RFC, Naive Bayes, Logistic Regression, LSTM, and KNN. Overall, all models have high precision, recall, and F1-scores. K-Nearest Neighbors would be the least recommended model for building a phishing email detection system, though. It also has the lowest F1-scores.

Classic ML Models					
	Accuracy	***Label***	***Precision***	***Recall***	***F1-score***
LR	0.9849	0	1.00	0.97	0.99
		1	0.97	1.00	0.98
SVM	0.9911	0	1.00	0.99	0.99
		1	0.99	1.00	0.99
Naive Bayes	0.9862	0	0.98	0.99	0.99
		1	0.99	0.98	0.99
RFC	0.9867	0	0.99	0.98	0.99
		1	0.98	0.99	0.99
KNN	0.9755	0	0.98	0.97	0.98
		1	0.97	0.98	0.97
Deep Learning					
	Accuracy	***Precision***	***Recall***	***F1-score***	
LSTM	0.9843	0.9806	0.9863	0.9834	
GRU	0.9870	0.9741	0.9989	0.9863	

Fig 3. Accuracy, Precision, Recall, and F1-score of ML and DL models

Fig. 4 shows the confusion matrix made from training the SVM model on the Enron dataset, because the SVM model has the highest accuracy out of all the machine learning and deep learning models trained. As shown by the figure, the true positives and true negatives are dark blue, while the false positives and negatives are nearly white. There is a high number of true positives and negatives compared to false positives and false negatives, so it can be concluded that the SVM model can accurately predict whether emails are phishing or non-phishing.

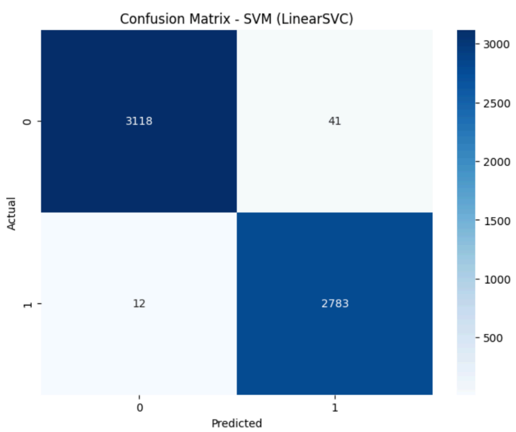

Fig.4. Confusion Matrix of SVM

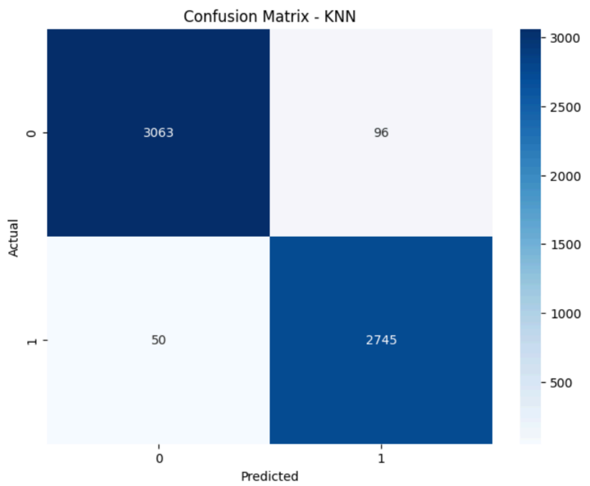

Fig.5. Confusion matrix of KNN

As shown in Fig. 5, the KNN model has a higher number of false negatives and false positives than the SVM model, especially in the false negatives. Even though the SVM model has a higher accuracy than the KNN model, both machine learning models have high true positives and true negatives according to their confusion matrices. From just looking over each confusion matrix, the human eye can barely tell the difference. This may be because there is not much of a difference in accuracy between KNN (0.9755) and SVM (0.9911) in real-life scenarios.

It can be concluded that the KNN model is still a reliable model for developing an automatic spam email detection system, but it is still not the best choice in comparison to the other machine learning and deep learning models trained and analyzed in this project.

VI. Future Work

Exploring and comparing different NLP (natural language processing) techniques to TF-IDF, such as Word2Vec, is a possible way to expand upon the current research in this paper. Another possible future direction to take is training more deep learning models, since there were only two deep learning models in this current project. A good choice would be the deep learning model BERT (Bidirectional Encoder Representations from Transformers), a model used for NLP. The number of epochs can be increased, too, when training the deep learning models.

Other than accuracy, precision, recall, and F1-score, metrics such as space and time complexity can also be explored. Space complexity can be measured by the amount of memory each model uses when trained, and time complexity can be measured by its speed or the amount of time it takes to train. Loss functions can also be analyzed in future work.

Since the Enron dataset only covered emails from Enron, future work can include an even larger dataset from a variety of companies. The Enron dataset only contains the subject and body data emails, too, but other components, such as the sender, can also be explored in future work. The NLP features can be explored and analyzed to see which words or text content are the most relevant in the machine learning models' predictions of whether an email is phishing or not.

VII. Conclusion

It can be concluded that machine learning models are effective in predicting phishing email attacks. By using Google Colab and Python libraries, 5 classic machine learning model classifiers, Logistic Regression (LR), SVM (Support Vector Machine), Naive Bayes (NB), RFC (Random Forest Classifier), KNN (K-Nearest Neighbors), and 2 recurrent neural networks (RNN), Long Short-Term Memory (LSTM), and Gated Recurrent Unit (GRU), were trained on a email dataset to find which models are the most reliable in developing a spam email detection system. The NLP technique TF-IDF (Term Frequency-Inverse Document Frequency) was utilized to create features for the models to train on.

Before training the data, word clouds were created to see if there were any obvious differences between phishing and non-phishing emails. Results from training were then collected and recorded in a table, and confusion matrices were created for further analysis.

Since the SVM model has the highest accuracy out of all the machine learning and deep learning models at 99.11%, it can be concluded as the best choice as a phishing email detection system. Its F1-scores for both labels are high too, at 99%, indicating a reliable model. Among the deep learning models trained in this project, the GRU model has a higher accuracy and has the second-highest accuracy out of all the machine learning models trained. It can be concluded that the KNN model would be the worst choice out of the models trained, because it has the lowest accuracy of 97.55%. KNN also has the lowest F1-score for label 1, which is 0.97.

With the results showing high accuracy in every model, machine learning models are an effective method in developing automatic phishing detection systems to protect the safety of Internet users from scams, fraud, and possible leaking of sensitive information. These cybersecurity applications can benefit many people in real-life scenarios, considering the prevalence of Internet use in daily life. Machine learning models for predicting phishing emails can also prevent companies from financial loss and be an efficient spam filter.

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